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Modified Newton-Type Iteration Methods for Generalized Absolute Value Equations

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Abstract

In this paper, by separating the differential and the non-differential parts of the generalized absolute value equations, a class of modified Newton-type iteration methods are proposed. The modified Newton-type iteration method involves the well-known Picard iteration method as the special case. Convergence properties of the new iteration schemes are analyzed in detail. In particular, some specific sufficient conditions are presented for two special coefficient matrices. Finally, two numerical examples are given to illustrate the effectiveness of the proposed modified Newton-type iteration methods.

Keywords Generalized absolute value equations · Newton method · Convergence · Differential function

Mathematics Subject Classification 65F10 · 90C05 · 90C30

1 Introduction

The generalized absolute value equations (GAVE) and the absolute value equations (AVE) arise in many areas of scientific computing and engineering applications, such as linear programming, bimatrix games, quadratic programming and so on [1,2]. The

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well-known linear complementarity problem (LCP) can be transformed into the GAVE or the AVE, and vice versa [3,4]. For details of the relation between the GAVE and the LCP, we refer the readers to [5–8].

Recently, the GAVE and the AVE have been attracted by many researchers. On the one hand, some scholars tried to present sufficient conditions to guarantee the existence of the solution for the GAVE and the AVE; see [5,9–12] for more detailed descriptions. On the other hand, some scholars have proposed many efficient iteration methods to solve the GAVE and the AVE. There are the concave minimization algorithm [13,14], the sign accord algorithm [15], the Picard and the Picard-HSS iteration methods [11, 16], the optimization algorithm [17], the hybrid algorithm [18] and so on.

By considering the AVE as a system of nonlinear equations and introducing the subgradient for the absolute value term, Mangasarian proposed a generalized Newton (GN) iteration method to solve the AVE [19]. Theoretical analysis showed that the GN iteration method is globally linear convergent to a solution of the AVE. Then, the GN iteration method and its convergence condition have been extended by Hu et al. [20] to solve the GAVE. Recently, convergence results of the GN iteration method have been further studied in [21–24]. From the GN iteration method, we can see that at each iteration step the Jacobian matrix is changed. For large problems, it is very difficult especially if the Jacobian matrix is ill-conditioned. To remedy this, several improvements have been proposed, such as a stable and quadratic convergent GN iteration scheme [24], the generalized Traub's method [25], the modified GN iteration method [26], the inexact semi-smooth Newton iteration method [27] and so on.

For a system of nonlinear equations with the nonlinear function being the sum of a differentiable function and a Lipschitz continuous function, a modified Newton-type (MN) iteration method has been studied in [28] by separating the differential part and the non-differential part. In this paper, the MN iteration method is applied to solve the GAVE and the AVE. With a special choice of the differential function, a specific scheme is given. The new scheme involves the well-known Picard iteration method [11] as the special case. A general sufficient condition to guarantee the convergence of the MN iteration method is proved. Moreover, some specific sufficient conditions are established when the coefficient matrix A is either a symmetric positive definite matrix or an H_+-matrix.

The rest of this paper is organized as follows. In Sect. 2, the modified Newton-type iteration method is introduced to solve the GAVE and the AVE. We also present the specific iteration schemes. In Sect. 3, convergence analysis of the modified Newton-type iteration method is studied in detail. In Sect. 4, two numerical examples are listed to show the effectiveness of the proposed method. Finally, we give some conclusions in Sect. 5.

2 Modified Newton-Type Iteration Method

As is known, the form of the GAVE is given, i.e.,

$$Ax - B|x| = b, \tag{1}$$

where $A, B \in \mathbb{R}^{n\times n}$, $b \in \mathbb{R}^n$ and $|\cdot|$ stands for the componentwise absolute value. In particular, if $B = I$, where I denotes the identity matrix, then the GAVE (1) reduces to the absolute value equations (AVE).

$$Ax - |x| = b. \tag{2}$$

Due to the existence of the nonlinear term $B|x|$, the GAVE (1) can be regarded as a system of nonlinear equations

$$F(x) = 0, \quad \text{with} \quad F(x) = Ax - B|x| - b. \tag{3}$$

Thus, the well-known Newton iteration method

$$x^{(k+1)} = x^{(k)} - F'(x^{(k)})^{-1} F(x^{(k)}), \quad k = 0, 1, 2, \dots \tag{4}$$

can be used provided that the Jacobian matrix $F'(x)$ of $F(x)$ exists and is invertible. However, the Newton iteration scheme (4) cannot be used directly to solve the GAVE (1) since $F(x) = Ax - B|x| - b$ is non-differentiable. For the special case, i.e., $B = I$, by considering $F(x) = Ax - |x| - b$ as a piecewise linear vector function, Mangasarian used the generalized Jacobian $\partial|x|$ of $|x|$ based on a subgradient of its components and proposed the following generalized Newton (GN) iteration method [19]

$$x^{(k+1)} = (A - D(x^{(k)}))^{-1} b, \quad k = 0, 1, 2, \dots \tag{5}$$

to solve the AVE (2), where $D(x^{(k)}) = \partial|x| = \mathrm{diag}(\mathrm{sign}(x^{(k)}))$ and $\mathrm{sign}(x)$ denotes a vector with components equal to 1, 0 or -1 depending on whether the corresponding component of x is positive, zero or negative. For the GAVE (1), the GN iteration scheme (5) becomes

$$x^{(k+1)} = (A - BD(x^{(k)}))^{-1} b, \quad k = 0, 1, 2, \dots. \tag{6}$$

For the system of nonlinear equations

$$F(x) = 0, \quad \text{with} \quad F(x) = H(x) + G(x), \tag{7}$$

where $F(x)$ is assumed to be a sum of a differentiable function $H(x)$ and a Lipschitz continuous function $G(x)$. By separating the differential part $H(x)$ and the non-differential part $G(x)$, Han [28] studied the following modified Newton-type (MN) iteration scheme to get an approximation solution of (7)

$$x^{(k+1)} = x^{(k)} - H^{'}(x^{(k)})^{-1}(H(x^{(k)}) + G(x^{(k)})), \quad k = 0, 1, 2, \dots \tag{8}$$

provided that the Jacobian matrix $H^{'}(x^{(k)})$ is invertible. Convergence conditions have been established. It is clear that the MN iteration method (8) reduces to the classical Newton iteration method (4) when $G(x) = 0$. Note that similar idea has been studied by Bai and Yang for solving a class of weakly nonlinear equations [29].

In the following, we will apply the MN iteration scheme (8) to solve the GAVE (1). In fact, the nonlinear function $F(x)$ in the GAVE (1) or (3) can be regarded as a sum of a differentiable function and a Lipschitz continuous function. By introducing a positive semi-definite matrix $\Omega \in \mathbb{R}^{n\times n}$, a simple and possible way is to let

$$H(x) = Ax + \Omega x \quad \text{and} \quad G(x) = -\Omega x - B|x| - b.$$

It is easy to check that the Jacobian matrix $H^{'}(x) = A + \Omega$. If $A + \Omega$ is nonsingular, then we can get a new iteration method, which is also named as the modified Newton-type (MN) iteration method

$$\begin{aligned} x^{(k+1)} &= x^{(k)} - (A+\Omega)^{-1}(Ax^{(k)} - B|x^{(k)}| - b) \\ &= (A+\Omega)^{-1}(\Omega x^{(k)} + B|x^{(k)}| + b), \quad k = 0, 1, 2, \ldots \end{aligned} \tag{9}$$

for solving the GAVE (1).

In particular, if Ω is the zero matrix, then the MN iteration method (9) reduces to

$$\begin{aligned} x^{(k+1)} &= x^{(k)} - A^{-1}(Ax^{(k)} - B|x^{(k)}| - b) \\ &= A^{-1}(B|x^{(k)}| + b), \quad k = 0, 1, 2, \ldots, \end{aligned} \tag{10}$$

which is known as the Picard iteration method [11]. So, our new iteration method is a generalization of the well-known Picard iteration method. Note that the MN iteration method (9) can be directly used to solve the AVE (2).

Comparing the MN iteration method (9) with the GN iteration method (6), we can see that at each iteration step of the MN iteration method, the coefficient matrix is unchanged, while at different iteration steps of the GN iteration method different systems of linear equations need to be solved. From this point of view, the MN iteration method (9) may have better computing efficiency than the GN iteration method (6). Although at each iteration step of the Picard iteration method the coefficient matrix is unchanged, $A + \Omega$ may be better conditioned than A due to the positive semi-definite matrix Ω. Thus, the MN iteration method may be more easily solved than the Picard iteration method, especially when inexact solvers are used.

3 Convergence Analysis

In this section, we study the convergence conditions for the MN iteration method (9). A general convergence condition is derived firstly. Then, some specific convergence conditions are derived when the matrix A is either symmetric positive definite or an H_+-matrix. As a special case of the MN iteration method, convergence conditions of the Picard iteration method can be obtained directly.

To this end, the following notations and definitions are used. A matrix is given as $A = (a_{ij}) \in \mathbb{R}^{n\times n}$. The matrix A is said to be a symmetric positive definite if A is symmetric and satisfies $x^T Ax > 0$ for all $x \in \mathbb{R}^n\backslash\{0\}$. If $a_{ij} < 0$ for any $i \neq j$, then A is called a Z-matrix. If A is a Z-matrix and $A^{-1} \geq 0$, we say that A is an M-matrix. The comparison matrix $\langle A\rangle = (\langle a\rangle_{ij}) \in \mathbb{R}^{n\times n}$ is defined as

$$\langle a\rangle_{ij}=\begin{cases}|a_{ij}|, & \text{for } i=j,\\ -|a_{ij}|, & \text{for } i\neq j,\end{cases}\quad i,j=1,2,\ldots,n.$$

If $\langle A\rangle$ is an M-matrix, we say that A is an H-matrix, and an H_+-matrix if it is an H-matrix and its diagonal entries are positive [30]. $\|A\|_2$ denotes the 2-norm of the matrix A.

3.1 General Sufficient Convergence Conditions

The following two theorems present sufficient conditions to guarantee the convergence of the MN iteration method (9) for solving the GAVE (1) when the matrices $A+\Omega$ and A are nonsingular, respectively.

Theorem 3.1 *Let $A, B\in\mathbb{R}^{n\times n}$ and $\Omega\in\mathbb{R}^{n\times n}$ be positive semi-definite such that $A+\Omega$ is invertible. If*

$$\|(A+\Omega)^{-1}\|_2<\frac{1}{\|\Omega\|_2+\|B\|_2},\tag{11}$$

then the MN iteration method (9) converges linearly from any starting point to a solution $x^{(*)}$ of the GAVE (1).

Proof Let $x^{(*)}$ be a solution of the GAVE (1), then it satisfies

$$Ax^{(*)}-B|x^{(*)}|=b.\tag{12}$$

Note that the MN iteration scheme (9) can be rewritten as

$$(A+\Omega)x^{(k+1)}=\Omega x^{(k)}+B|x^{(k)}|+b.\tag{13}$$

Subtracting (12) from (13), we obtain

$$\begin{aligned}A(x^{(k+1)}-x^{(*)})&=-\Omega(x^{(k+1)}-x^{(k)})+B(|x^{(k)}|-|x^{(*)}|)\\&=-\Omega(x^{(k+1)}-x^{(*)}+x^{(*)}-x^{(k)})+B(|x^{(k)}|-|x^{(*)}|)\\&=-\Omega(x^{(k+1)}-x^{(*)})+\Omega(x^{(k)}-x^{(*)})+B(|x^{(k)}|-|x^{(*)}|).\end{aligned}$$

Hence,

$$(A+\Omega)(x^{(k+1)}-x^{(*)})=\Omega(x^{(k)}-x^{(*)})+B(|x^{(k)}|-|x^{(*)}|).$$

Since $A+\Omega$ is nonsingular, we have

$$x^{(k+1)}-x^{(*)}=(A+\Omega)^{-1}(\Omega(x^{(k)}-x^{(*)})+B(|x^{(k)}|-|x^{(*)}|)).\tag{14}$$

Taking the 2-norm on both sides of (14), we have

$$\|x^{(k+1)}-x^{(*)}\|_2\leq\|(A+\Omega)^{-1}\|_2(\|\Omega\|_2+\|B\|_2)\|x^{(k)}-x^{(*)}\|_2,\tag{15}$$

where the inequality $\| \, |x^{(k)}| - |x^{(*)}| \, \|_2 \le \|x^{(k)} - x^{(*)}\|_2$ is used. From (15), we can see that the MN iteration method (9) converges linearly from any starting point to a solution $x^{(*)}$ of the GAVE (1) provided that condition (11) is satisfied. □

Theorem 3.2 *Let $A \in \mathbb{R}^{n\times n}$ be invertible, $B \in \mathbb{R}^{n\times n}$, and $\Omega \in \mathbb{R}^{n\times n}$ be positive semi-definite such that $A + \Omega$ is invertible. If*

$$\|A^{-1}\|_2 < \frac{1}{2\|\Omega\|_2 + \|B\|_2}, \tag{16}$$

then the MN iteration method (9) converges linearly from any starting point to a solution $x^{(*)}$ of the GAVE (1).

Proof According to the proof of Theorem 3.1, we can see from (15) that if

$$\|(A+\Omega)^{-1}\|_2(\|\Omega\|_2 + \|B\|_2) < 1,$$

then the MN iteration method (9) is linearly convergent. Under condition (16), by the Banach perturbation Lemma [31, Lemma 2.3.3] we have

$$\begin{aligned} \|(A+\Omega)^{-1}\|_2 &\le \frac{\|A^{-1}\|_2}{1 - \|A^{-1}\|_2 \cdot \|\Omega\|_2} \\ &< \frac{\frac{1}{2\|\Omega\|_2+\|B\|_2}}{1 - \frac{\|\Omega\|_2}{2\|\Omega\|_2+\|B\|_2}} \\ &= \frac{1}{\|\Omega\|_2 + \|B\|_2}. \end{aligned}$$

Therefore, the MN iteration method (9) converges linearly from any starting point to a solution $x^{(*)}$ of the GAVE (1) provided that condition (16) is satisfied. □

Since the AVE (2) is a special case of the GAVE (1). The MN iteration method (9) can be directly used to solve the AVE (2). By simply letting $B = I$, the following corollary can be obtained.

Corollary 3.1 *Let $A \in \mathbb{R}^{n\times n}$. Let $\Omega \in \mathbb{R}^{n\times n}$ be positive semi-definite such that $A+\Omega$ is invertible. If*

$$\|(A+\Omega)^{-1}\|_2 < \frac{1}{1 + \|\Omega\|_2},$$

or if A is invertible and

$$\|A^{-1}\|_2 < \frac{1}{1 + 2\|\Omega\|_2},$$

then the MN iteration method (9) converges linearly from any starting point to a solution $x^{(*)}$ of the AVE (2).

In Sect. 2, we have shown that our new MN iteration method (9) is a generalization of the well-known Picard iteration method (10). Let Ω be the zero matrix, we have the following two corollaries.

Corollary 3.2 *Let $A \in \mathbb{R}^{n\times n}$ be invertible and $B \in \mathbb{R}^{n\times n}$. If*

$$\|A^{-1}\|_2 < \frac{1}{\|B\|_2},$$

then the Picard iteration method (10) *converges linearly from any starting point to a solution $x^{(*)}$ of the GAVE* (1).

Corollary 3.3 *Let $A \in \mathbb{R}^{n\times n}$ be invertible. If*

$$\|A^{-1}\|_2 < 1,$$

then the Picard iteration method (10) *converges linearly from any starting point to a solution $x^{(*)}$ of the AVE* (2).

3.2 The Case of Symmetric Positive Definite Matrix

In this subsection, we discuss the convergence conditions of the MN iteration method (9) for solving the GAVE (1) and the AVE (2) when the matrix A is symmetric positive definite and $\Omega = \omega I$ is a positive scalar matrix.

Theorem 3.3 *Let $A \in \mathbb{R}^{n\times n}$ be a symmetric positive definite matrix, $\Omega = \omega I \in \mathbb{R}^{n\times n}$ with $\omega > 0$. Let $\mu_{\min}$ be the smallest eigenvalue of the matrix A and $\|B\|_2 = \tau$. If*

$$\tau < \mu_{\min}, \tag{17}$$

then the MN iteration method (9) *converges linearly from any starting point to a solution $x^{(*)}$ of the GAVE* (1).

Proof According to the proof of Theorem 3.1, we just need to get the sufficient conditions for $\|(A+\Omega)^{-1}\|_2(\|\Omega\|_2 + \|B\|_2) < 1$.

By assumptions, we have

$$\begin{aligned}\|(A+\Omega)^{-1}\|_2(\|\Omega\|_2 + \|B\|_2) &= \|(A+\omega I)^{-1}\|_2(\|\omega I\|_2 + \|B\|_2)\\ &= \frac{\omega+\tau}{\omega+\mu_{\min}}\end{aligned}$$

and $\|(A+\Omega)^{-1}\|_2 < \frac{1}{\|\Omega\|_2+\|B\|_2}$ provided that $\tau < \mu_{\min}$ (17) holds. This completes the proof. □

In particular, if $B = I \in \mathbb{R}^{n\times n}$, the following corollary can be obtained.

Corollary 3.4 *Let $A \in \mathbb{R}^{n\times n}$ be a symmetric positive definite matrix and $\Omega = \omega I \in \mathbb{R}^{n\times n}$ with $\omega > 0$. Let $\mu_{\min}$ be the smallest eigenvalue of the matrix A. If*

$$1 < \mu_{\min},$$

then the MN iteration method (9) converges linearly from any starting point to a solution $x^{(*)}$ of the AVE (2).

From Theorem 3.3 and Corollary 3.4, we can see that the convergence conditions of the MN iteration method (9) are independent of the additional positive scalar matrix Ω. This means that the convergence conditions presented in Theorem 3.3 and Corollary 3.4 are suitable for the Picard iteration method (10) for solving the GAVE (1) and the AVE (2), respectively. However, thanks to the presence of the positive scalar matrix Ω, the matrix $A + \Omega$ can be expected to be better conditioned than the matrix A. In addition, $A + \Omega$ can be strictly diagonally dominant matrix. Thus, our new MN iteration may have better computing efficiency than the well-known Picard iteration method.

3.3 The Case of H_+-Matrix

Theorem 3.4 *Let $A \in \mathbb{R}^{n\times n}$ be an H_+-matrix, $B \in \mathbb{R}^{n\times n}$ and $\Omega \in \mathbb{R}^{n\times n}$ be a positive scalar matrix. If*

$$\|(\Omega + \langle A\rangle)^{-1}\|_2 < \frac{1}{\|\Omega + |B|\,\|_2}, \tag{18}$$

then the MN iteration method (9) converges linearly from any starting point to a solution $x^{(*)}$ of the GAVE (1).

Proof By assumptions and [30, Lemma 3.4], it holds that

$$|(A + \Omega)^{-1}| \le (\langle A\rangle + \Omega)^{-1}.$$

Taking absolute value on both sides (14), we have

$$\begin{aligned} |x^{(k+1)} - x^{(*)}| &\le |(A+\Omega)^{-1}| \cdot |\Omega| \cdot |x^{(k)} - x^{(*)}| + |(A+\Omega)^{-1}| \cdot |B| \cdot |x^{(k)} - x^{(*)}| \\ &\le (\langle A\rangle + \Omega)^{-1} \cdot (\Omega + |B|)|x^{(k)} - x^{(*)}| \\ &:= Z|x^{(k)} - x^{(*)}|. \end{aligned} \tag{19}$$

Evidently, the iteration sequence $\{x^{(k)}\}_{k=0}^{+\infty}$ converges to a solution $x^{(*)}$ if the spectral radius of the matrix Z is less than one [6], i.e., $\rho(Z) < 1$. According to the definition of the matrix Z (19), we have

$$\rho(Z) \le \|(\Omega + \langle A\rangle)^{-1}(\Omega + |B|)\|_2 \le \|(\Omega + \langle A\rangle)^{-1}\|_2 \cdot \|\Omega + |B|\,\|_2.$$

Therefore, if condition (18) holds, then the MN iteration method (9) converges linearly from any starting point to a solution $x^{(*)}$ of the GAVE (1). $\square$

If $B = I$, then the convergence condition of the MN iteration method (9) for solving the AVE (2) can be obtained directly. Furthermore, the convergence conditions of the Picard iteration method (10) can be obtained for solving the GAVE (1) and the AVE (2). These theoretical results are summarized in the following three corollaries.

Corollary 3.5 *Let $A \in \mathbb{R}^{n\times n}$ be an H_+-matrix and $\Omega \in \mathbb{R}^{n\times n}$ be a positive scalar matrix. If*

$$\|(\Omega + \langle A\rangle)^{-1}\|_2 < \frac{1}{\|\Omega + I\|_2},$$

then the MN iteration method (9) *converges linearly from any starting point to a solution $x^{(*)}$ of the AVE* (2).

Corollary 3.6 *Let $A \in \mathbb{R}^{n\times n}$ be an H_+-matrix and $B \in \mathbb{R}^{n\times n}$. If*

$$\|\langle A\rangle^{-1}\|_2 < \frac{1}{\|\,|B|\,\|_2},$$

then the Picard iteration method (10) *converges linearly from any starting point to a solution $x^{(*)}$ of the GAVE* (1).

Corollary 3.7 *Let $A \in \mathbb{R}^{n\times n}$ be an H_+-matrix. If*

$$\|\langle A\rangle^{-1}\|_2 < 1,$$

then the Picard iteration method (10) *converges linearly from any starting point to a solution $x^{(*)}$ of the AVE* (2).

4 Numerical Experiments

In this section, we present two numerical examples to show the effectiveness and robustness of the MN iteration method (9) and show the advantages of the MN iteration method over the GN iteration method (6) [19], the modified generalized Newton (MGN) iteration method [26] and the Picard iteration method (10) [11] from aspects of the iteration counts (denoted by "IT") and the elapsed CPU times (denoted by "CPU"). Here, the specific scheme of the MGN iteration method is defined as

$$x^{(k+1)} = (A + I - BD(x^{(k)}))^{-1}(x^{(k)} + b), \quad k = 0, 1, 2, \ldots.$$

In our experiments, the initial guess vector is $x^{(0)} = (1, 0, \ldots, 1, 0, \ldots)^T$. All runs are terminated once

$$\mathrm{RES}(x^{(k)}) := \frac{\|Ax^{(k)} - B|x^{(k)}| - b\|_2}{\|b\|_2} \leq 10^{-7}$$

or if the prescribed iteration number $k_{\max} = 5000$ is exceeded. For simplicity, we choose $\Omega = \omega I$ in our MN iteration method. Thus, at each iteration step of the MN

Table 1 Numerical comparison of the testing methods for Example 4.1

Method	n			
	1000	2000	3000	4000
GN				
IT	2	2	2	2
CPU	0.52	2.62	6.71	13.61
RES	8.09e−16	1.15e−15	1.49e−15	1.57e−15
MGN				
IT	2	2	2	2
CPU	0.64	2.86	7.61	15.73
RES	3.71e−09	4.65e−10	1.37e−10	5.83e−11
Picard				
IT	2	2	2	2
CPU	0.25	1.22	3.27	6.47
RES	1.85e−09	2.33e−10	6.86e−11	2.91e−11
MN				
ω_{exp}	0.8	0.8	0.8	0.7
IT	2	2	2	2
CPU	0.24	1.20	3.19	6.39
RES	6.00e−09	7.52e−10	2.22e−10	8.42e−11

iteration method, the GN iteration method, the MGN iteration method and the Picard iteration method, we need to solve a system of linear equations with the coefficient matrix $A + \omega I$, $A - BD(x^{(k)})$, $A + I - BD(x^{(k)})$ and A, respectively. Here, these systems of linear equations are solved by the sparse LU factorization (for nonsymmetric case) or the sparse Cholesky factorization (for symmetric positive definite case) in combination with an approximate minimum degree (AMD) or column AMD reordering. To get fast convergence rate of the MN iteration method, the parameter ω is chosen the experimentally optimal one ω_{exp}, which leads to the smallest iteration step. All computations are run in MATLAB (version R2014a) in double precision, Intel(R) Core(TM) (i5-3337U CPU, 8G RAM) Windows 8 system.

Example 4.1 The first example belongs to the AVE (2). The matrix A is randomly generated by the following MATLAB procedure:

$$A = R^T * R + n * \text{eye}(n) \quad \text{with} \quad R = rand(n, n),$$

and all singular values of the matrix A exceed 1. The right-hand side is chosen as $b = (A - \text{eye}(n, n)) * \text{ones}(n, 1)$.

For the first example, we choose four increasing sizes, i.e., $n = 1000, 2000, 3000$ and 4000. Note that the randomly generated matrices are dense. Numerical results of the different iteration methods are listed in Table 1. From Table 1, we can see that the iteration steps of the discussed four iteration methods are the same while the elapsed CPU times are different. Compared with the GN and the MGN iteration methods, the

MN iteration method and the Picard iteration method have better computing efficiency. By computation, we know that the main costs are the sparse LU factorization of the system matrix A. For the GN and the MGN iteration methods, the sparse LU factorization should be done twice, while for the MN and the Picard iteration methods, only one sparse LU factorization needs to be computed. Therefore, from aspects of the elapsed CPU times, we can see that the proposed MN iteration method and the well-known Picard iteration method are much better than the GN iteration method and the MGN iteration method for solving the AVE (2).

Example 4.2 ([6]) The second example comes from the LCP: For a given matrix $M \in \mathbb{R}^{n\times n}$ and a given vector $q \in \mathbb{R}^n$, find $z, w \in \mathbb{R}^n$ such that

$$z \geq 0, \quad w = Mz + q \geq 0 \quad \text{and} \quad z^T w = 0. \tag{20}$$

From [4–6], we can see that the LCP (20) is equivalent to

$$(M+I)x - (M-I)|x| = q \quad \text{with} \quad x = \frac{1}{2}((M-I)z + q), \tag{21}$$

which belongs to the GAVE. Here, we take $A = M + I$ and $B = M - I$, in which $M \in \mathbb{R}^{n\times n}$ is given by $M = \widehat{M} + \mu I \in \mathbb{R}^{n\times n}$ and $q \in \mathbb{R}^n$ is given by $q = -Mz^{(*)}$, where

$$\widehat{M} = \mathrm{Tridiag}(-I, S, -I) = \begin{bmatrix} S & -I & 0 & \cdots & 0 & 0 \\ -I & S & -I & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -I & S \end{bmatrix} \in \mathbb{R}^{n\times n}$$

is a block tridiagonal matrix,

$$S = \mathrm{Tridiag}(-1, 4, -1) = \begin{bmatrix} 4 & -1 & 0 & \cdots & 0 & 0 \\ -1 & 4 & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 4 \end{bmatrix} \in \mathbb{R}^{m\times m}$$

is a tridiagonal matrix, $n = m^2$ and $z^{(*)} = (1.2, 1.2, 1.2, \ldots, 1.2, \ldots)^T \in \mathbb{R}^n$ is the unique solution of the LCP (20). The exact solution of the GAVE (21) is $x^{(*)} = (-0.6, -0.6, -0.6, \ldots, -0.6, \ldots)^T \in \mathbb{R}^n$.

To further show the advantages of our proposed MN iteration method (9), we choose three cases for the parameter μ, i.e., $\mu = 4, \ -1, \ -4$. By computation, we know that the matrices A and M are symmetric positive definite when $\mu = 4$, and symmetric indefinite when $\mu = -4$. For the case $\mu = -1$, the matrix M is symmetric indefinite while A is symmetric positive definite. For each case, we choose five increasing sizes of n, i.e., $n = 3600, \ 4900, \ 6400, \ 8100, \ 10{,}000$. Note that the matrices A and B are sparse for this example. Recently, for the LCP, one of the most useful iteration method

Table 2 Numerical comparison of the testing methods for Example 4.2 with $\mu = 4$

Method	n				
	3600	4900	6400	8100	10,000
GN					
IT	2	2	2	2	2
CPU	0.16	0.28	0.46	0.77	1.16
RES	5.45e−16	5.65e−16	5.97e−16	6.15e−16	6.14e−16
MGN					
IT	9	9	9	9	9
CPU	0.85	1.52	2.48	3.86	5.64
RES	1.95e−08	1.96e−08	1.98e−08	1.99e−08	1.99e−08
Picard					
IT	77	76	76	76	75
CPU	0.09	0.12	0.16	0.21	0.25
RES	8.62e−08	9.60e−08	9.02e−08	8.54e−08	9.72e−08
MN					
ω_{exp}	5.1	5.1	5.1	5.1	5.1
IT	12	12	12	12	12
CPU	0.04	0.05	0.06	0.08	0.10
RES	4.92e−08	4.98e−08	5.03e−08	5.07e−08	5.11e−08

is the modulus iteration method [4]. From the iteration scheme studied in [4], we can find that the modulus iteration method is exactly the Picard iteration method (10).

In Table 2, we list the numerical results of the discussed four iteration methods for $\mu = 4$. For this example, we can see from Table 2 that in terms of iteration counts, the GN iteration method is the best one and the Picard iteration method is the worst one. The iteration steps of the MGN iteration method and the MN iteration method are more than those of the GN iteration method. However, from the aspect of the elapsed CPU times, we can see that our new proposed MN iteration method has the best computing efficiency since it cost much cheaper than the other three iteration methods.

In Tables 3 and 4, the numerical results of the discussed four iteration methods are presented for $\mu = -1$ and $\mu = -4$, respectively. In these two tables, "–" means that the corresponding iteration method does not converge to the approximate solution within $k_{\max}$ iteration steps or even diverge. As discussed before, when $\mu = -1$, the matrix A is symmetric positive definite while the matrix M is symmetric indefinite. So, it may satisfy $\|(I+M)^{-1}(I-M)\|_2 \geq 1$. Thus, the convergence of the Picard iteration method (i.e., the modulus method in [4]) cannot be guaranteed. The same reason can be obtained for $\mu = -4$. In addition, when $\mu = -4$, both the matrices A and M are symmetric indefinite. From the numerical results presented in Table 4, we can see that both the GN iteration method and the Picard iteration converge slowly or even diverge, while the MGN iteration method and our new proposed MN iteration method converge fast. Although the iteration steps show that the MGN iteration method converges faster than the MN iteration method, the elapsed CPU times show that our new proposed

Table 3 Numerical comparison of the testing methods for Example 4.2 with $\mu = -1$

Method	n				
	3600	4900	6400	8100	10,000
GN					
IT	16	16	15	15	15
CPU	1.06	1.96	3.11	4.92	7.66
RES	5.08e−08	4.33e−08	9.89e−08	8.76e−08	7.87e−08
MGN					
IT	16	16	16	16	16
CPU	1.13	2.15	3.61	5.92	8.68
RES	8.45e−08	7.46e−08	6.76e−08	6.23e−08	5.83e−08
Picard					
IT	–	–	–	–	–
CPU	–	–	–	–	–
RES	–	–	–	–	–
MN					
$\omega_{\exp}$	1.2	1.2	1.2	1.2	1.2
IT	45	45	44	44	44
CPU	0.07	0.08	0.11	0.14	0.18
RES	8.35e−08	7.76e−08	9.50e−08	8.97e−08	8.53e−08

Table 4 Numerical comparison of the testing methods for Example 4.2 with $\mu = -4$

Method	n				
	3600	4900	6400	8100	10,000
GN					
IT	–	–	–	–	–
CPU	–	–	–	–	–
RES	–	–	–	–	–
MGN					
IT	22	22	22	23	23
CPU	1.62	2.91	4.92	8.18	12.61
RES	5.58e−08	7.55e−08	9.82e−08	4.13e−08	5.08e−08
Picard					
IT	–	–	–	–	–
CPU	–	–	–	–	–
RES	–	–	–	–	–
MN					
$\omega_{\exp}$	4.2	4.2	4.2	4.2	4.2
IT	42	42	42	41	41
CPU	0.06	0.08	0.10	0.14	0.16
RES	8.80e−08	8.23e−08	7.77e−08	9.78e−08	9.40e−08

MN iteration method has better computing efficiency. Moreover, this example shows that the new MN iteration method may be suitable for solving the GAVE (1) when A is indefinite although we did not present any theoretical results. Most importantly, Tables 2, 3 and 4 show that the MN iteration method (9) is the "optimal" one since the iteration steps are independent of the problem sizes. Therefore, our new proposed MN iteration method is very effective.

5 Conclusions

In this paper, by separating the differential and the non-differential parts of the GAVE (1), a class of modified Newton-type iteration methods are introduced. With a special choice of the differential function, a specific iteration scheme is presented. The new iteration scheme can be regarded as a generalization of the well-known Picard iteration method [11]. But we can adopt suitable differential function such that the new iteration method has better convergence rate and better computing efficiency. Convergence properties of the proposed MN iteration method are analyzed in detail. Numerical results show that the new iteration method is powerful for solving the GAVE (1).

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